arXiv:2408.12757v2 [cs.DC] 25 May 2025

NanoFlow: Towards Optimal Large Language Model Serving Throughput

Kan Zhu
University of Washington

Yufei Gao
University of Washington
Tsinghua University

Yilong Zhao
University of Washington
UC Berkeley

Liangyu Zhao
University of Washington

Gefei Zuo
University of Michigan

Yile Gu
University of Washington

Dedong Xie
University of Washington

Tian Tang
University of Washington
Tsinghua University

Qinyu Xu
University of Washington
Tsinghua University

Zihao Ye
University of Washington

Keisuke Kamahori
University of Washington

Chien-Yu Lin
University of Washington

Ziren Wang
University of Washington
Tsinghua University

Stephanie Wang
University of Washington

Arvind Krishnamurthy
University of Washington

Baris Kasikci
University of Washington

Abstract

Large Language Models (LLMs) have resulted in a surging demand for planet-scale serving systems, where tens of thousands of GPUs continuously serve hundreds of millions of users. Consequently, throughput has emerged as a key metric that determines serving systems' performance. Due to large model sizes and memory-intensive self-attention, LLM serving has been commonly assumed to be memory-bound. Through a detailed analysis, we show that despite having memory-intensive components, end-to-end LLM serving is compute bound for most common workloads and LLMs. Alas, most existing serving engines fall short from optimal compute utilization, because the heterogeneous operations that comprise LLM serving—compute, memory, networking—are executed sequentially within a device.

We propose NanoFlow, a novel serving framework that exploits intra-device parallelism, which overlaps the usage of heterogeneous resources within a single device. NanoFlow splits inputs into smaller *nano-batches* and duplicates operations to operate on each portion independently, enabling overlapping. NanoFlow automatically identifies the number, size, ordering, and GPU resource allocation of nano-batches to minimize the execution time, while considering the interference of concurrent operations. We evaluate NanoFlow's end-to-end serving throughput on several popular models such as LLaMA-2-70B, Mixtral 8×7B, LLaMA-3-8B, etc. With practical workloads, NanoFlow provides 1.91× throughput boost compared to state-of-the-art serving systems, achieving between 50% to 72% of optimal throughput across popular models.

1 Introduction

Large language models (LLMs) based on transformers power applications like chatbots, search engines, and office software but are highly resource-intensive [21, 28, 41]. Reports indicate over 200 million weekly ChatGPT users, with API usage doubling after GPT-4o Mini's release [31, 35, 40]. Given the massive demand and limited GPU availability [11, 47], maximizing hardware utilization is critical. In particular, throughput, measured as tokens per device per second, has become a key factor in reducing serving costs [13, 17, 26].

LLMs are more resource-intensive than earlier DNNs due to unique characteristics. Their model sizes are vastly larger, such as GPT-3 with 175B parameters requiring 5×A100 80 GB GPUs to store weights in 16-bit precision, with some closed-source models even larger [7]. Moreover, LLMs' self-attention mechanism [51] scales total memory loading quadratically with the context length—the aggregate length of input and output sequences—during token generation. LLM serving systems often cache per-request state in a *key-value cache* (KV-cache), whose size can surpass the size of model weights [44], driving up memory demands. Moreover, each LLM iteration loads the entire model weights plus the KV-cache, while only producing one output token for each de-

coding sequence, further increasing memory pressure. Due to these characteristics, LLM serving is commonly assumed to be overall memory bandwidth bound.

In this paper, we show that while the self-attention operation is indeed memory-bound, *LLM serving is often compute-bound when considered as a whole*. We provide a detailed analysis (which we also validate empirically) of the common operations used in current LLMs and find that: (1) Batching prefill and decode requests amortizes weight loading, making general matrix multiplications (GEMMs) compute-bound.(2) Although batched decode request still need to load a unique KV-cache per sequence, novel optimizations such as grouped query attention [5] (GQA) can reduce the memory loads. (3) As model sizes grow, the compute operations from GEMMs tend to dominate compared to self-attention. Consequently, for many common workloads and LLMs, the total compute operations dominate network and memory I/O.

However, in practice, we find that LLM serving engines are far from optimal in terms of compute utilization when measured end-to-end and compared to the hardware FLOPs. This is because LLMs use *heterogeneous operations*. For example, the self-attention is memory-bound during the decode phase. Also, when models are larger than a single GPU's memory capacity, GEMM is often split across multiple GPUs when using tensor parallelism, and thus requires network communication in addition to compute. We find that in current LLM serving engines, individual operations have high utilization of their bottlenecked resource ($\sim 80\%$). However, because each operation is executed one at a time, and different operations have different resource bottlenecks, the total compute utilization is only 40%, as shown in Section 6.2.

To address this gap, we present NanoFlow, a serving framework that aims to maximize utilization of the workload's resource bottleneck, when considered as a whole. The key insight is to leverage *intra-device parallelism for heterogeneous operations*. In particular, NanoFlow splits each input batch into *nano-batches* and duplicates operations across nano-operations, with each nano-operation processing a single nano-batch. Since nano-operations operate on separate nano-batches without dependencies, heterogeneous operations—such as memory-bound and compute-bound operations—can execute *simultaneously* rather than sequentially. This approach facilitates fine-grained pipelining across compute, memory, and network resources within a *single device*. As nano-operations duplicate each operation, this method increases memory I/O for weight loading. However, when the workload is compute-bound as a whole, the additional memory I/O can be hidden via pipelining.

To achieve efficient pipelining, NanoFlow executes nano-operations with a fraction of the GPU's resources. For instance, while most of the GPU resources can be assigned to a compute-heavy operation for high compute utilization, a memory-bound operation can still achieve considerable memory bandwidth usage using the remaining GPU resources.

Designing NanoFlow presents several major challenges. First, given the multitude and diversity of LLMs, determining the appropriate number of nano-batches, the input batch size per each nano-operation, and the execution order of nano-operations involves navigating a large search space. Second, when memory- or network-bound kernels run alongside compute-bound kernels, competition for GPU resources (e.g., execution units, caches) causes unpredictable performance interference, slowing down kernels running in parallel [43]. Balancing these trade-offs is critical. Allocating more resources to compute-bound kernels may excessively delay memory- or network-bound kernels, while prioritizing memory- or network-bound kernels could hinder overall throughput by slowing down compute-bound operations.

To tackle these challenges and efficiently explore the large search space, NanoFlow proposes *auto-search* to automatically construct an intra-device pipeline. The auto-search process reduces the search space by executing in two stages, which approximates the optimal solution balancing efficiency and accuracy. In the first stage, NanoFlow determines the initial pipeline schedule, including the number, size, and ordering of nano-operations, assuming no interference between kernels. In the second stage, NanoFlow refines the pipeline by profiling actual kernel interference and re-planning accordingly. NanoFlow also provides a runtime system to execute the optimized pipeline by efficiently forming nano-batches, allocating GPU resources, and managing the KV-cache.

We provide a detailed evaluation of NanoFlow on the LLaMA-2-70B model [49] using one NVIDIA 8×A100 DGX node [24]. For practical workloads including ShareGPT [1], LMSys [56] and Splitwise [32], NanoFlow achieves on average 1.91× greater throughput compared with state-of-the-art serving frameworks including vLLM [17], DeepSpeed-FastGen [13] and TensorRT-LLM [26], which is 68.5% of the theoretically optimal throughput we derive in §3. NanoFlow achieves similar latency compared with the best baseline TensorRT-LLM at a low request rate, while handling up to 1.64× higher request rates within SLO constraint. We also demonstrate NanoFlow's ability to automatically generate efficient pipelines for other models and architectures such as Mixture-of-Experts (MoE), by evaluating popular LLMs (LLaMA-3-70B, LLaMA-3-8B, QWen2-72B, Deepseek-67B, and Mixtral 8×7B). NanoFlow-generated serving pipelines achieve 50%-72% of optimal throughput, which on average reaches 2.66× throughput gain compared with vLLM.

In summary, we contribute the following:

- A detailed analysis and empirical validation of the workload characteristics and theoretically optimal throughput of LLM serving systems that shows modern LLMs operate in a compute-bound regime.
- NanoFlow, that comprises (1) an auto-search engine that automatically computes a pipeline for nano-batches to improve LLM serving throughput, (2) an end-to-end LLM serving runtime.

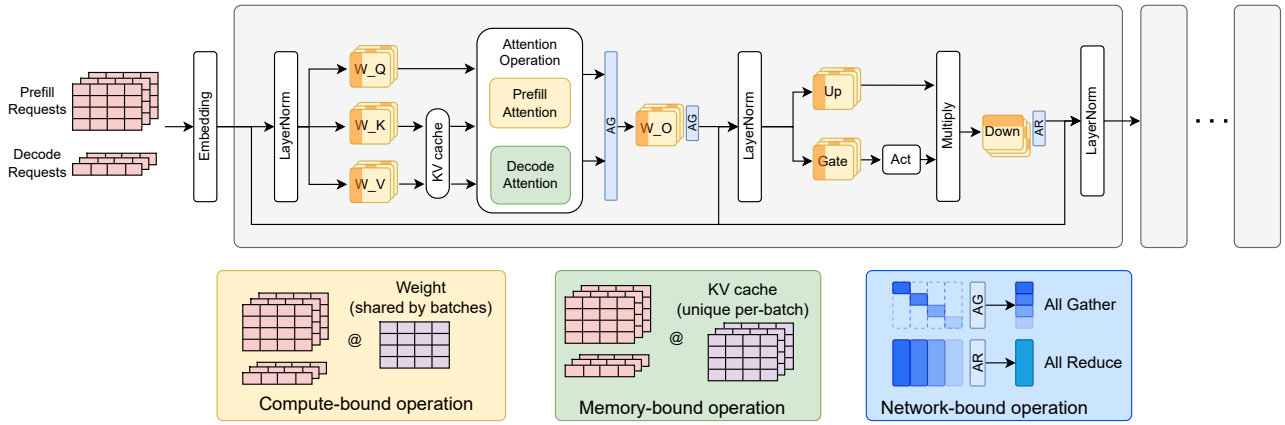


Figure 1: Transformer architecture. The operations in the yellow boxes have large batch sizes and share model weight parameters across requests; hence, they are compute-bound. Operations in green boxes require loading a unique KV cache for each request; hence, they are memory-bound. The blue box represents network operations that perform synchronization between operations.

- A comprehensive evaluation of NanoFlow that achieves 1.91× throughput gain compared to baselines and 68.5% of theoretical maximum throughput.

2 Background

2.1 LLM Inference Workflow

Recent LLMs like GPT-4, LLaMA, and Mistral [16, 29, 49] are based on the decoder-only transformer [51]. The inference process of these LLMs has two phases [53]: (1) *prefill*, which processes the input prompt all at once, and (2) *decode*, which generates output tokens one at a time auto-regressively. The prefill phase initializes the *KV-cache* [34], which keeps the per-request state to speed up the decode phase.

Both the prefill and decode phases utilize the same inference workflow in Figure 1. Upon receiving an inference request, the input traverses identical *decoder layers* to produce the subsequent tokens. In each decoder layer, during the attention stage, the input is multiplied by weight matrices W_Q, W_K, and W_V to form Query, Key, and Value, with Key and Value concatenated into the existing KV-cache. The Query is then multiplied with the existing Keys and normalized to assess the similarity between the current token and all previous tokens. This similarity is employed to compute the weighted average of the Value, aggregating the context information. The outcome undergoes O-projection through a linear transformation using W_O followed by a layer normalization operation [6]. Subsequently, during the feed forward network stage, the activation is multiplied by W_{up} and W_{gate} separately to generate o_u and o_g. Next, o_g passes through an activation function (e.g., SiLU [10]), followed by an element-wise multiplication with o_u. Finally, W_{down} is applied as Down-projection to generate the output, which serves as the input to the next layer.

2.2 Operation Characteristics

According to the characteristics of the operations during inference, we can classify them into four categories.

Dense operations. The KQV generation, O projection, Up, Gate, and Down projection compute the multiplication of activation and model weights, which we define as *dense operations*. In the prefill phase, all tokens of new requests form a large batch for these dense operations. In the decode phase, batched decoding aggregates the activations from all the decode requests for dense operations [53]. Moreover, as the prefill phase and decode phase share weight matrices, the activations from both phases can be combined to further exploit the batching effect by amortizing the weight loading cost [3]. As a result, dense operations are *compute-bound*.

Attention operations. Transformers capture token relations through attention [51]. The prefill phase processes all input tokens simultaneously, making it *compute-bound*, while the decode phase generates one token per iteration, loading cached K and V vectors from the KV-cache, making it *memory-bound*.

Network operations. Collective communication operations including AllGather (AG) and AllReduce (AR) are *network-bound* and call for high bandwidth interconnection between nodes like NVLink [25].

Other operations. We classify the rest of the operations in transformer architecture, e.g., layer norms, position embeddings, etc., as other operations, due to their short runtime compared to dense or attention operations.

2.3 Serving LLMs at Scale

With large models, multiple GPUs are required to provide enough resources, including both memory and compute, for efficient LLM serving [18]. Existing works [12, 38, 57] have comprehensive evaluations of different *inter-device paral-*

lelism paradigms:

Tensor Parallelism [38, 57] splits each weight matrix onto different GPUs and executes each operation collectively, avoiding weight duplication, thereby scaling compute capacity. However, frequent collective communications after operations are required to synchronize each operation's results, as shown in Figure 1.

Pipeline Parallelism [14] splits the model into stages by layers so that each GPU only needs to hold part of the weights. While all devices with tensor parallelism execute on the same batch of data, devices with pipeline parallelism operate on different micro-batches of data split by stages.

In practice, these parallelism paradigms are used in combination. Tensor parallelism is widely used on GPUs within a node to enable a larger batch size, while pipeline parallelism is used across nodes to further scale the cluster.

3 Analysis

We now introduce the factors determining the throughput of LLM serving (§3.1) and model the cost of LLM serving (§3.2). We then classify modern LLMs serving workloads according to our model (§3.3) and empirically validate this classification (§3.4), showing that popular workloads are compute-bound. Then, for the compute-bound scenarios, we determine optimal serving throughput (§3.5), explain why existing systems fall short of the optimal (§3.6), and how this motivates our proposed approach, intra-device parallelism (§3.7).

3.1 Key Factors of Serving Throughput

We define throughput as the *total throughput* of serving, which is the number of tokens processed per second by both prefill and decode phases. Throughput is affected by hardware properties, model configuration, user queries, and batch size.

Hardware specification. The following factors determine the available GPU hardware resources:

- N_{GPU}: number of GPUs
- $MemBW$ (GB/s): aggregate GPU memory bandwidth
- $MemSize$ (GB): aggregate GPU memory capacity
- $Compute$ (GFLOP/s): aggregate GPU compute capacity
- $NetBW$ (GB/s): aggregate GPU interconnect bandwidth

Model configuration. The following factors regarding the model architecture determine the computation, memory, and network demand when serving LLMs:

- D_{model}: hidden dimension size
- L: number of layers
- P_{Model}: number of parameters
- R_{GQA}: group size of GQA [1]
- S_{type} (Bytes): number of bytes (size) of the data type for model parameters (e.g., 2 for FP16)

User query statistics. The following user input statistics also affect the system throughput:

- p: average number of tokens in prompts to be prefilled
- d: average number of tokens in output to be decoded

[1] The number of attention heads with a shared KV head [5].

Therefore, a serving request from a user corresponds to p prefill tokens and d decode tokens on average, for a total of $p+d$ tokens that we use when calculating the total throughput. Using user query statistics, total throughput (tokens generated per second) can be easily converted to other throughput metrics, for example, decoding throughput (decode output per second) is $\frac{d}{p+d} \times \text{Throughput}_{\text{total}}$, and request per second is $\frac{1}{p+d} \times \text{Throughput}_{\text{total}}$.

Batch size. An optimal throughput-oriented serving system needs to process large batches. For compute-bound dense operations, e.g., GEMMs, a larger batch amortizes the weight loading overhead and increases execution unit occupancy, which provides higher compute utilization. For memory- and network-bound operations, all requests share the same warmup cost (e.g., kernel launching overhead, pipeline setup overhead, network synchronization overhead, etc.), so a larger batch size leads to higher throughput. Therefore, when considering the optimal system throughput, we assume that the system always operates at the largest batch size at which the total available memory can hold the model weights and all the KV caches for the requests. [2]

3.2 Cost Model of LLM Serving

Under the assumption of maximum batch size, we derive the latency of an LLM serving iteration (where a batch of user requests are being processed by all the transformer layers) from the perspectives of required memory, compute, and network resources.

Memory. From the memory perspective, latency is:

$$T_{mem} = \frac{MemSize}{MemBW} \tag{1}$$

This is because the entire device memory content needs to be loaded into the GPU caches and registers once for each iteration (typical in modern serving systems [17]) when using the largest batch size. Although the model weights are shared between iterations, due to the long reuse distance, it is infeasible to cache the model weights and avoid loading from the device memory.

Compute. In LLM serving, all operations contribute to the total compute, while Dense operations produce the vast majority of computations (as we validate in §3.4). Therefore we model the compute latency of dense operations below[3].

For every GEMM in dense operations, $2B_{Dense}N_wK_w$ computations need to be performed on weights and activations, where N_w and K_w are dimensions of the weight matrices, and B_{Dense} is the batch size for dense operations, which includes decode tokens from hundreds of requests and the prefill tokens from one or a few requests. Thus, the total compute for dense operations is $2B_{Dense}\sum N_wK_w$. Note that the $\sum N_wK_w$ term is the number of weight elements in all layers, which can

[2] Activations occupy negligible (less than 5%) of the memory [17], thus we omit them for simplicity.

[3] Our supplementary material provides a more detailed analytical model.

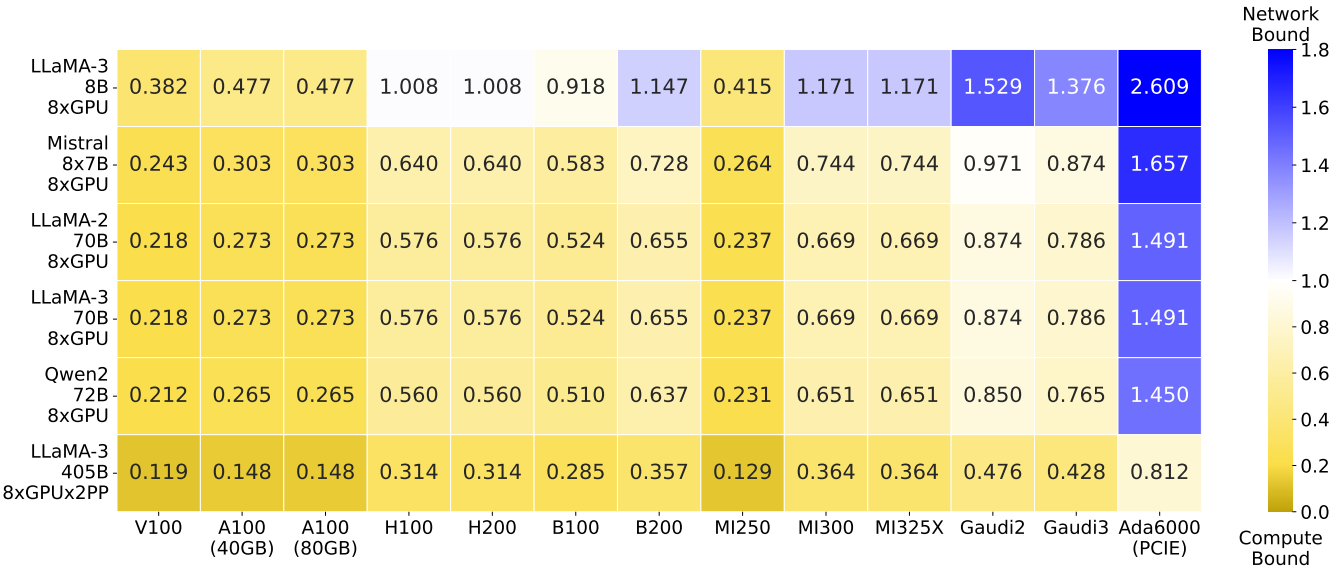


Figure 2: Comparison of network time and compute time. The closer to yellow, the more compute-bound the workload is, whereas the closer to blue indicates the workload is more network-bound.

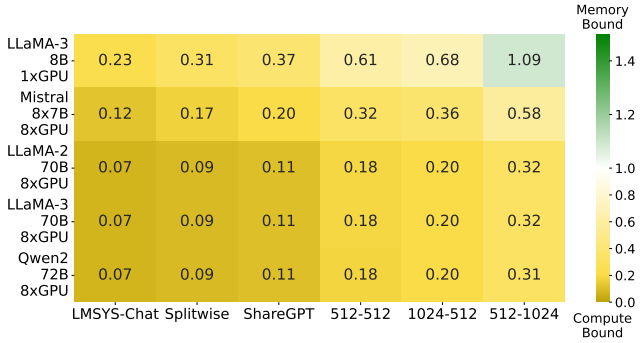


Figure 3: Comparison of compute time and memory time. The closer to yellow, the more compute-bound the workload is, whereas the closer to green the more memory-bound it becomes.

be approximated using P_{Model} (the parameters in the model), thus, we simplify the expression as $2B_{Dense}P_{Model}$. Therefore, latency from the compute perspective is:

$$T_{Compute} \approx \frac{2B_{Dense} \cdot P_{Model}}{Compute} \tag{2}$$

Network. Finally, for tensor parallelism, collective network communication primitives are needed to synchronize the results after performing operations in multiple GPUs over different shards of data. The size of the data transmitted with collective communication equals the size of inputs for dense operations, which have dimension $[B_{dense}, D_{model}]$. To support tensor parallelism, two AGs and one ARs (or 2 ARs) are required [39]. An AR requires gathering outputs from other GPUs, performing local summation, and broadcasting results, thus it needs to transfer the activations twice, while an AG transfers activations once. Therefore, in both cases, we can estimate the total data movement for one GPU in bytes as $4 \cdot B_{Dense}D_{model}S_{type} \cdot L$. Thus, the latency from the network perspective is:

$$T_{net} \approx 4 \cdot \frac{N_{GPU}B_{Dense}D_{model}S_{type}L}{NetBW} \tag{3}$$

3.3 Classification of LLM Serving Workloads

We classify LLM serving workloads by comparing the latency of an iteration derived based on memory, compute, and network requirements.

First, we compare Equation 3 (network time) with Equation 2 (compute time). Their ratio can be calculated as

$$\frac{T_{Net}}{T_{Compute}} = \frac{2D_{model}L}{P_{model}} \frac{N_{GPU}Compute}{NetBW/S_{type}}$$

Note that $P_{model} \approx 12D_{model}^2L$[4], thus $\frac{2D_{model}L}{P_{model}} \approx 1/(6D_{model})$. For models that need multiple GPUs (where T_{Net} is relevant), D_{model} is typically larger than 4096 [9, 16, 22, 48], making the term $\frac{2D_{model}L}{P_{model}}$ smaller than 1×10^{-5}. Due to the high bandwidth interconnect between data center GPUs (e.g., NVLink [25], Infinity Fabric [54]), $\frac{N_{GPU}Compute}{NetBW/S_{type}}$ ranges from 1×10^4 to 1×10^5. Therefore, this ratio is typically less than 1, indicating that the network is not

[4] $P_{model} = ((W_K + W_Q + W_V) + W_O + W_U + W_G + W_D)L \approx ((1/R_{GQA} + 1 + 1/R_{GQA})D_{model}^2 + D_{model}^2 + D_{model}I_{model} + D_{model}I_{model} + I_{model}D_{model})L) \approx (1 + 1 + 3.5 + 3.5 + 3.5)D_{model}^2L \approx 12D_{model}^2L$, where I_{model} is intermediate dimension, which is typically 3.5x of D_{model} [16, 22].

Table 1: Characteristics of accelerator models across various vendors and release years.

Vendor	Model	Release Year	MemSize (GB)	MemBW (GB/s)	NetBW (GB/s)	Compute (FP16 GFLOP/s)	$\frac{MemSize}{MemBW}$	$\frac{Compute}{MemBW}$	$\frac{NetBW}{MemBW}$
NVIDIA	V100	2017	16	900	300	125,000	0.018	139	0.33
NVIDIA	A100	2020	40	1,555	600	312,000	0.026	200	0.39
NVIDIA	A100	2021	80	2,000	600	312,000	0.040	156	0.30
NVIDIA	H100	2023	80	3,352	900	989,000	0.024	295	0.268
NVIDIA	H200	2024	96	4,800	900	989,000	0.020	206	0.19
NVIDIA	B100	2024	120	8,000	1,800	1,800,000	0.015	225	0.23
NVIDIA	B200	2024	120	8,000	1,800	2,250,000	0.015	281	0.23
AMD	MI250	2021	128	3,352	800	362,000	0.038	107	0.24
AMD	MI300	2023	192	5,300	1,024	1,307,000	0.036	246	0.19
AMD	MI325X	2024	256	6,000	1,024	1,307,000	0.043	218	0.17
Intel	Gaudi 2	2022	96	2,400	600	1,000,000	0.040	417	0.25
Intel	Gaudi 3	2024	128	3,700	1,200	1,800,000	0.035	486	0.32
NVIDIA	Ada 6000	2022	48	960	64	182,000	0.050	190	0.067

the bottleneck. We illustrate this ratio in Figure 2. For large models and data center GPUs, the compute takes more time than network.

We then compare Equation 1 (memory time) and Equation 2 (compute time) to finally determine the characteristics of the workload, namely, the following value

$$T_R = \frac{T_{Mem}}{T_{Compute}} \approx \frac{Compute}{MemBW} \frac{MemSize}{P_{model}} \frac{1}{2B_{dense}} \tag{4}$$

Modern models widely adopt Grouped Query Attention (GQA) as seen in LLaMA-3 [48], NeMo [4], and Qwen2 [9]. In GQA, multiple attention heads share a KV-cache, which effectively means that given the same memory capacity, batches can contain more requests. Consequently, B_{dense} tends to be larger. For example, when serving LLaMA-2 $70B$ on $8 \times A100$ GPUs, the maximum batch size of decode requests is on the order of 1024. Combined with the prefill tokens in the batch, B_{dense} can reach 2048 while a non-GQA model with the same size only can have $B_{dense} = 256$. Thus, the ratio T_R is less than 1, and compute becomes the main bottleneck. Additionally, modern models tend to have increasingly large model sizes P_{model} (whereas the rest of the terms in T_R are constants given a hardware specification), further causing T_R to drop below 1.

Figure 3 shows the values of T_R with five representative models: LLaMA-3 8B [22] with 1×A100, Mistral 8×7B [16] with 8×A100, LLaMA-2 70B [49] with 8×A100s, LLaMA-3 70B with 8×A100s, Qwen-2 72B [9] with 8×A100s. As the heatmap figure shows, many of the workloads (Splitwise [32], LMSYS-Chat-1M [56], ShareGPT [1] as well as constant length inputs and outputs), are uniformly compute-bound, except in the case of the long decode (512-1024) workload on LLaMA-3 8B model, where the ratio T_R is around 1.

While the analysis is mainly conducted on an NVIDIA A100, the result remains similar for other accelerators. As shown in Table 1, compute/memory ratio and compute/network ratio are stable across vendors and generations, meaning the workload characteristics also remain largely unchanged.

To summarize, our cost model suggests that modern LLM serving workloads operate in a compute-bound regime, which we validate empirically in the next section.

3.4 Validation of the Cost Model

We validated our cost model on LLaMA-2 70B serving with 8 NVIDIA A100 GPUs and a dense batch size of 2048 requests. We computed the GFLOPs, memory movement, and network traffic for each operation (Table 2) and used these with our model to calculate $T_{compute}$, T_{mem}, and T_{net} based on hardware specifications[5]. The longest estimated time $T_{op} = \max(T_{compute}, T_{mem}, T_{net})$ indicates the most constrained resource. To validate these results, we measure the running time of different operations. For most operations, T_{op} aligns with profiling results. An exception is prefill attention, where the real measured time is larger due to the launching overhead of the associated kernels dominating the total time. The sums of all operations' $T_{compute}$, T_{mem}, and T_{net} values show that compute is the most constrained resource, which aligns with our model's finding.

3.5 Optimal Serving Throughput

Optimal throughput (measured in token/s/GPU) is achieved when the most limited resource, i.e. compute, is fully utilized. By applying Equation 2, the system's optimal throughput is:

$$\text{Throughput}_{\text{optimal}} = \frac{B_{Dense}}{T_{Compute}} = \frac{Compute}{2P_{Model}} \tag{5}$$

In the compute-bound regime, the optimal throughput depends *solely* on the aggregate computational capacity of the GPUs for the specific data type and the number of parameters in the model. Notably, other factors, including the GPU memory size, bandwidth, model data type, or the length of prefill and decode phases, do not have a significant impact on achieving optimal throughput. We measure the GPU compute

[5] One-way network bandwidth was used for T_{net}

Table 2: Comparison of operation runtimes between cost model estimation and real-world measurements.

Operation	Compute (GFLOP)	Mem Load (GB)	Net Usage (GB)	Est. T_{comp} (ms)	Est. T_{mem} (ms)	Est. T_{net} (ms)	Real Time (ms)
KQV	27487.8	19.5	0	**11.01**	1.22	0	**16.08**
O	21990.2	16.1	0	**8.81**	1.01	0	**16.01**
UG	153931.6	96.6	0	**61.67**	6.04	0	**69.92**
D	76965.8	49.7	0	**30.84**	3.11	0	**34.96**
DecAttn	3665.9	462.2	0	1.47	**28.89**	0	**35.60**
PfAttn	916.3	2.1	0	**0.37**	0.13	0	**4.56**
Net	18.8	75.2	75.2	0.01	4.70	**31.33**	**47.92**
Total				**114.17**	45.09	31.33	

capacity by profiling the state-of-the-art GEMM vendor library, CUTLASS [46]. For example, when serving LLaMA-2 70B on 8×A100 GPUs, the profiled peak $Compute$ is 280 TFLOPS for FP16, and the P_{Model} is 70B. Substituting these into Equation 5, yields to an optimal throughput of 1857 tokens/s/GPU.

3.6 Gap to Optimal Throughput

As shown in Figure 7, existing serving systems fall far short of achieving optimal throughput. When serving offline requests, vLLM, DeepSpeed-FastGen, and TensorRT-LLM achieve only 22.0%, 22.9%, and 37.8% of the optimal throughput, respectively.

The main reason why existing serving systems fall significantly short of achieving optimal throughput is poor utilization of GPU resources. Figure 4 shows the execution flow of transformer operations within a GPU, using existing serving frameworks like SGLang [58], vLLM [17], DeepSpeed-FastGen [13]. These frameworks execute compute bound operations, decode attention, and network operations sequentially on a single large batch in a device. This sequential execution flow forms multiple pipeline bubbles (denoted as "WASTED" in Figure 4), which cause the most constrained resource, compute, to be underutilized.

3.7 Intra-device Parallelism

Ostensibly, the advantage of existing serving frameworks when using a single large batch in a GPU is that they need to load model weights fewer times than if there were multiple smaller batches and corresponding operations.

However, the cost model and its validation that we outlined earlier in this section provides a valuable insight: Since modern models are largely compute-bound (by a factor of more than 2 as shown in Figure 3 and Table 2), creating and processing smaller batches may be justified, especially if compute utilization can be increased.

These observations motivate intra-device parallelism through nano-batching, which underlies the design of NanoFlow. NanoFlow creates multiple nano-operations, which are the same operations as the original operations, but operate on finer-grained batches that we call nano-batches. For example, for the original Up projection operating on a batch size of 2048, NanoFlow could create 2 nano-operations UP1 and UP2, operating on batches in ranges 0-768 and 768-2048, respectively, while performing the exact same projection. Different nano-batches do not have data dependencies. Therefore, nano-operations that use different resources—compute, memory, network—can operate on different nano-batches in parallel within a device to maximize the utilization of the most constrained resource, compute. Despite requiring repeated loading of model weights, we demonstrate that NanoFlow provides significantly more throughput than existing frameworks (1.91× on average), further closing the gap to optimal serving throughput (§6).

4 Design

We introduce NanoFlow's key components: the auto-search engine for automatic nano-batch pipeline computation (§4.1) and the LLM serving runtime for pipeline execution (§4.2).

4.1 Automated Pipeline Search

Deciding each nano-operation's resource allocation, launching kernels, and scheduling with intra-device parallelism is challenging due to model and hardware diversity. To address this, NanoFlow proposes auto-search, which uses mixed integer linear programming to obtain the nano-batch overlapping strategy, with a two-stage approximation for reducing the search space. We first outline auto-search prerequisites including kernel profiling and interference modeling (§4.1.1), then detail its two stages: pipeline structure search (§4.1.2) and GPU resource allocation (§4.1.3). Finally, we provide example pipelines for popular models (§4.1.4).

4.1.1 Kernel Profiling and Interference Modeling

Understanding the performance characteristics of different operations is key to automated pipeline search. For this, NanoFlow profiles GEMM kernels (used for compute-bound operations such as KQV generation), GEMV kernels (used in memory-bound operations as in decode-attention), and network kernels (used in network-bound operations such as AR and AG).

Determining the maximum dense batch size Given a specific model architecture and user input statistics (as described in §3.1), NanoFlow calculates the maximum *dense batch size* that can fit into the GPU memory (e.g., 2048 for $8 \times A100$ serving LLaMA-2 70B). This dense batch size serves as an input to auto-search, which sets an upper bound of shapes in profiling.

Profiling interference-free kernels NanoFlow then determines the interference-free execution time of kernels, where each operation exclusively uses the GPU. NanoFlow profiles kernels with discrete input batch sizes from 128 to the dense batch size in multiples of 128, as 128 is a hardware-friendly shape for GEMM tiling. NanoFlow explores all possible kernel implementations varying the number of thread blocks, the number of warps, and tile size for GEMM, GEMV, and network kernels to identify the implementation that provides

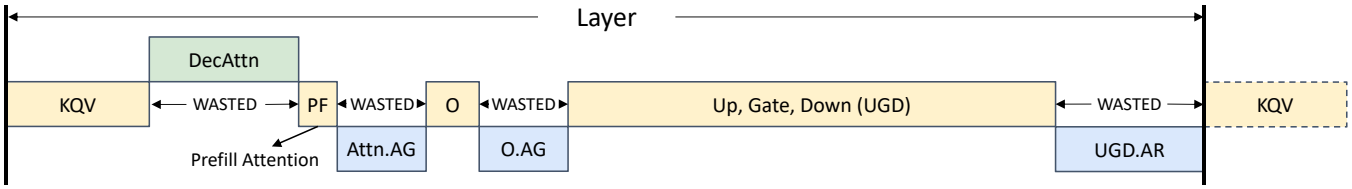


Figure 4: Execution pipeline of existing systems. The green, yellow, and blue operations correspond to memory-, compute-, and network-bound operations. Operations in the previous and next layer are denoted by dotted borders. "WASTED" shows the stages in the pipeline where the most constrained resource, compute, is underutilized. Small operations (i.e. layernorm, activation, etc.) are omitted for simplicity.

the shortest execution time. The profiling outputs a mapping from (kernel, batch size) to its best implementation.

Profiling kernels with interference Ideally, we would like to independently assign each nano-operation a fractional allocation of a single GPU's compute, memory, and network bandwidth. We call this fraction $R_{physical}$. However, when two or more kernels execute in parallel on a device, they slow down compared to running separately. We refer to this as *kernel interference*, which is attributed to kernels competing for GPU hardware resources such as execution units, caches, and memory bandwidth [43]. Unfortunately, kernel interference is unpredictable, as NVIDIA GPUs do not give explicit control over compute, memory, and network bandwidth. Thus, we cannot directly control $R_{physical}$.

Instead, we use GEMM performance as a proxy R for $R_{physical}$ and model pairwise kernel interference. We define R in a GEMM-centric way since the compute-bound operations dominate and optimizing compute is the goal of NanoFlow.

In particular, we profile pairwise kernel interference patterns by measuring actual performance when overlapping between a compute kernel A and another kernel B. For example, if when overlapped, kernel A achieves 40% performance compared to its best performance when run individually, then we say that R_A=0.4. We then assume that the overlapped kernel is given the remaining resources, so we set $R_B = 1 - R_A$ =0.6.

Then, we model the interference by measuring the performance loss in the non-compute kernel B when R_A resources are allocated to kernel A. When overlapping A and B, we normalize A and B's performance to their peak performance when run individually. We call this value P. For the compute kernel A, P_A is equivalent to R_A by definition, while P_B captures the memory or network utilization when R_A resources are assigned to the compute kernel.

Intuitively, this process establishes an exchange rate between R compute utilization to P memory or network utilization. The overall kernel interference model can then be captured in a table such as Table 3, which maps R to P for each kernel type.

Nonetheless, profiling the mapping between R and P remains challenging due to the vast number of possible kernel combinations. While each operation may have only a few hundred kernel implementations, the combinations between different implementations of overlapping kernels exponentially expand the profiling space, resulting in millions of possible configurations. This immense complexity makes exhaustive exploration infeasible.

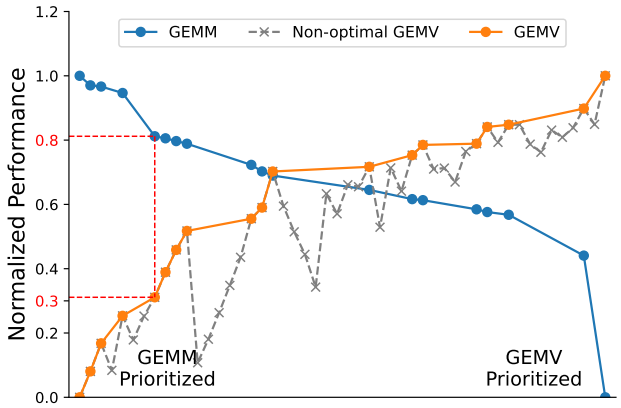


Figure 5: Interference characteristics between GEMM and GEMV kernels. The points on the x-axis correspond unique GEMM-GEMV implementation pairs. The y-axis denotes the GEMM and GEMV kernels' normalized performance P.

NanoFlow reduces the profiling space by limiting the thread block numbers for GEMV and network kernels to values between 8 and 128 (in steps of 8), as 128 blocks are sufficient to saturate their performance. It also excludes less efficient GEMM kernels which have longer execution times while using more thread blocks. NanoFlow focuses on pairwise interference analysis—compute-memory and compute-network—rather than examining three kernel types simultaneously and assumes that the R to P mapping profiled with pairwise interference holds when overlapping three kernels.

Figure 5 shows NanoFlow's profiling results of GEMM-GEMV kernel pairs ($\sim$ 100 pairs, after the aforementioned simplifications) on an A100 GPU with GEMM shape $(M,N,K) = (384,4096,4096)$, GEMV batch size 384, and sequence length 1024. These pairs are then sorted by descending GEMM performance, as shown in Figure 5. For instance, a normalized performance of 0.8 means the kernel's execution time is $1/0.8$ times longer than the fastest implementation.

Table 3: Performance P of GEMV and network kernels with different resource utilization R.

Operations	Resource Utilization (R)						
	0	0.1	0.2	...	0.8	0.9	1
GEMM (by definition)	0	0.1	0.2	...	0.8	0.9	1
GEMV	0	0.2	0.3	...	0.85	0.95	1
Network	0	0.3	0.5	...	0.9	1	1

Pairs with higher interference on GEMM but worse GEMV performance (grayed-out in Figure 5) are discarded. This process identifies the best-performing kernel combinations at various GEMM-GEMV performance trade-off points.

Using interference profiles, auto-search generates an *resource mapping table* (Table 3). This table quantifies GEMV and network kernel performance (P) as functions of resource utilization (R). For example, as depicted by the dotted red line in Figure 5, achieving 0.3 GEMV performance requires sacrificing 0.2 GEMM performance (from 1 to 0.8). This implies a non-linear trade-off: 0.2 GEMM performance is exchanged for 0.3 GEMV performance, indicating that $R_{\text{GEMV}} = 0.2$ corresponds to $P_{\text{GEMV}} = 0.3$.

A sensitivity analysis of all GEMM shapes in the evaluated LLM models and 64 batch size combinations reveals that the R to P mapping is consistent, with a standard deviation within 5% of the mean. Consequently, Table 3 can be used to map R to P for all GEMV and network kernel implementations in subsequent auto-search analysis.

4.1.2 Auto-search Stage I: Pipeline Structure Search

In the first stage, auto-search takes as input the dense batch size, the operation dependencies (defined in the PyTorch implementation of a given model architecture), and the profiling of interference-free kernels, and then produces as output, the number, batch size, and order of each nano-operation using mixed integer linear programming (MILP). Stage I does not factor in the interference between nano-operations to simplify the problem, which is covered by Stage II. The MILP problem is subject to the constraints described below.

Optimization objective. The main objective of MILP is to remove pipeline bubbles (denoted by wasted in Figure 4) for compute operations to minimize execution time.

Constraints on the number of nano-operations. To overlap resources, each operation needs to be split into at least two nano-operations, each operating on an independent nano-batch. As splitting into nano-operations reduces the batching effect, the number of nano-operations should be minimized. Thus, auto-search begins its search by dividing all operations into two nano-operations and formulating the MILP problem to identify the batch size, execution time, and ordering of the nano-operations. If the temporary best solution has bubbles for compute, auto-search increases the number of nano-operations for operations near the bubble to improve resource utilization until MILP cannot produce better solutions.

Constraints on batch sizes and execution times. Auto-search forces the batch sizes to be chosen from 128 to dense batch size and uses the interference-free execution time of a nano-operation for a given batch size, as computed in §4.1.1.

Constraints on dependencies. The dependencies of nano-operations are determined by their parent operations (present in baseline implementations, e.g., in pytorch) and their input batches. Two nano-operations are dependent if and only if their parent operations are dependent (e.g., an O-projection depends on an attention operation) and their input batches intersect (e.g., ranges 0–255 and 128–383).

Constraints on overlapping. Overlapping kernels constrained by the same resource (e.g., compute) is unhelpful for utilization, so auto-search restricts the pipeline to overlap only operations constrained by different resources.

Constraints on operation transformations. Some network collectives have equivalent transformations with different performance characteristics and dependencies. For example, an AG can be converted into an AR with different weight partitioning approaches [39, 57]. NanoFlow explores all of these alternative transformations for network nano-operations and searches for the shortest pipeline design.

Due to the large search space, searching for a optimal pipeline would take hours or even days. However, a practical pipeline can be found in approximately 10 minutes by prioritizing searching a feasible solution instead of finding a provably-optimal solution.

4.1.3 Auto-search Stage II: Refining the Pipeline

The pipeline derived in the first stage does not factor in kernel interference, which is impractical for real deployment. In the second stage, auto-search refines this pipeline by considering kernel slowdown due to interference. This stage formulates a separate MILP problem, where the numbers, batch sizes, and ordering constraints of nano-operations remain the same (i.e., outputs of the first stage). Auto-search explores various resource utilization (R) allocations to individual nano-operations, and maps R to P using Table 3 derived in interference profiling. A specific resource utilization value R directly corresponds to kernel implementations as determined in §4.1.1. The goal of the second stage is again to minimize pipeline execution time.

Constraints on GPU resource utilization. As discussed in §4.1.1, concurrent kernels should compete for in total of 1.0 of GPU resource utilization R. Thus, auto-search forces the sum of R at any given time to be less than or equal to 1.0.

Constrains on execution times. Given the resource utilization R for a nano-operation, auto-search computes the execution time of the operation using D_{best}/P, where D_{best} represents the best execution time without interference and P is derived using Table 3.

The auto-search is performed only when the model architecture or workload (input length, output length) undergoes

significant changes. Compared to the long-running deployment times, the search time is negligible.

4.1.4 Example Pipelines

70B pipeline We apply auto-search to 70B-scale models, including LLaMA-2 70B, LLaMA-3 70B, Qwen2.5-72B, and Deepseek-67B. Compared to LLaMA-2 70B, LLaMA-3 70B features a larger vocabulary size of 128K, which increases the sampling time; Deepseek-67B has a slightly different number of layers and hidden dimension (D_{model}), Qwen2.5-72B introduces bias in KQV generation. However, none of these adjustments significantly change the performance characteristics, leading to similar schedules. We provide the NanoFlow-generated pipeline for LLaMA-2 70B as an example in Figure 6. Since three resources (compute, memory, and network) overlap at the beginning of a decoding layer (i.e., KQV generations), auto-search ends up using 4 nano-operations for this part, to reduce pipeline bubbles. Here, the decode attention operation has resource utilization 0.4 (i.e. reduces 40% GEMM performance) and reaches 80% of maximum decode attention performance. For the remainder of the pipeline, GEMM operations are prioritized, and only two nano-operations are used.

8B pipeline 8B models (e.g. Llama3-8B) do not require network operations as they fit in a single GPU. Thus auto-search splits all operations into two nano-operations, with decode attention overlapping with the Up Gate Down projection.

MoE pipeline Compared to 70B models, MoE models have different hidden dimensions and number of layers. Due to the imbalance between experts [16], NanoFlow uses tensor parallelism for MoEs, in which FFN is implemented using grouped-GEMM. FFN also involves an additional gate routing operation. Auto-search works similarly for an MoE model and automatically produces an efficient pipeline (§6).

4.2 NanoFlow Runtime

We now explain how NanoFlow executes the auto-generated pipeline by asynchronous scheduling (§4.2.1), and KV-cache SSD offloading for multi-round conversations (§4.2.2).

4.2.1 Request Scheduling

Batch formation. NanoFlow assumes that auto-scaling, workload balancing, and priority-aware routing are managed by the control plane externally [15,33,45]. A NanoFlow instance assumes an abundance of requests and treats each request with equal priority. When requests are not abundant, the control plane should reduce the number of NanoFlow instances to maintain a sufficiently large per-instance batch size. Note that while the B_{dense}, i.e. the token batch size for GEMMs, is on the order of thousands (e.g., 1 prefill request of 2048 tokens and 256 decode tokens), this represents the combination of prefill and decode tokens. Therefore, the request batch size (e.g., 1 prefill and 256 decode requests) is on the order of hundreds, which is easy to attain in real-world large-scale serving.

When forming a batch, NanoFlow prioritizes unfinished decode requests and chunks prefill requests at a token granularity following SarathiServe [3] to exactly fill the remaining capacity of a pre-selected best-performing dense batch. This effectively reduces the tail latency as dense operations are performed over consistent batch sizes across iterations. Initially, the global batch contains only prefill requests (requests in prefill stages). As these requests complete the prefill stage, they transition to decode requests within the batch, and new prefill requests are introduced to maintain the fixed token batch size. As decode requests increase, the prefill token budget decreases, causing prefill requests to slow down. When decode requests begin completing, this frees space for more prefill tokens again, speeding up prefill. Thus, the number of decode requests and prefill requests will be stabilized.

To optimize GPU memory usage and avoid running out of memory, NanoFlow predicts peak future memory usage based on the status of user requests. NanoFlow tracks decoded tokens per request, estimates completion time using average decode length, and prefills new requests only if predicted memory usage stays within GPU limits. If the GPU runs out of memory, NanoFlow offloads a request to the CPU and reloads it once memory is available without recomputation.

Asynchronous scheduling. Batch formation, including estimating memory usage, scheduling new requests, retiring finished requests, and adjusting the page table for PagedAttention [17], consumes a non-negligible amount of time on the CPU side [42]. In most serving frameworks [17,58], *only after* the GPU executes one iteration, the scheduler on the CPU is able to detect *EOS* tokens, remove the finished request from the batch, and refill the batch with new requests. However, GPU is under-utilized during this time. To avoid this waste, NanoFlow asynchronously schedules batch formation in parallel to the GPU executions. At any iteration i, NanoFlow forms the batch for the next iteration *before* the end of the current iteration. This means that NanoFlow cannot detect the *EOS* tokens generated at iteration i. After launching iteration $i+1$, NanoFlow forms the batch for cycle $i+2$, detects the *EOS* token from iteration i, and removes finished requests. Fortunately, since the average decode length surpasses 100 for typical workloads (See Table 4), the overhead of one extra decode token is negligible ($< 1\%$), given the benefit of hiding the batch formation overhead.

4.2.2 KV-cache Management

To efficiently serve multi-round applications, NanoFlow offloads the KV-cache of running requests to a hierarchical cache consisting of host CPU memory and SSDs, and loads a prior round's KV-cache when a new round arrives.

Simultaneous offloading. Instead of waiting for requests' completion, NanoFlow offloads the KV-cache of tokens directly after the KQV generation in each transformer layer before appending them to the KV-cache. The KV vectors output by KQV generation are naturally contiguous and the

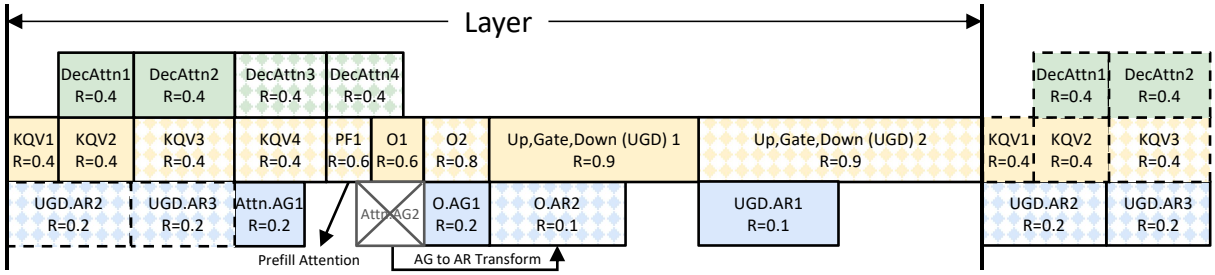


Figure 6: Execution pipeline of LLaMA-2 70B, automatically generated by NanoFlow. The solid background and shaded background represents input batch 0-768 and 768-2048, respectively. *R* stands for resource utilization. By overlapping the compute-, memory-, and network-intensive operations, NanoFlow increases compute utilization and improves the serving throughput.

data size of the offload is balanced across iterations by design as NanoFlow's dense batches remain stable. As a result, the host and the GPU hold the same copy of on-the-fly requests' KV-cache. To minimize the offloading overhead, NanoFlow performs the device-host copy—a GPU-initiated copy operation that uses minor GPU resources—during compute-bound operations in FFN. Moreover, NanoFlow reduces offloading time via NUMA-aware thread-binding.

Host KV-cache management. NanoFlow uses the LRU policy to manage the hierarchical cache of CPU memory and SSDs. For e.g., KV-cache are evicted to SSDs when the CPU reaches its memory limit and retrieved from either CPU or SSD to initiate the loading process when the next round conversation of a request comes in.

KV-cache loading and scattering. NanoFlow uses PagedAttention [17], therefore, the pages of the KV-cache reside in the GPU memory in a fragmented manner. In order to avoid copying to fragmented GPU page destinations, NanoFlow first copies the data to a contiguous space on GPU and then scatters the pages to their destinations. This achieves 7-10× higher bandwidth for host-to-device copy.

5 Implementation

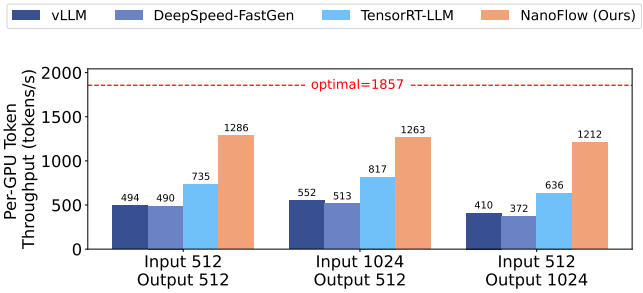
We implement NanoFlow for NVIDIA GPUs. NanoFlow consists ∼10K lines of CUDA and ∼6K lines of Python.

Given GPU resource requirements determined by auto-search, NanoFlow knows which kernels to launch for each nano-batch to achieve a given utilization level, based on its profiling results. NanoFlow launches nano-operations respecting the ordering constraints from auto-search and enforces ordering dependencies using CUDA events.

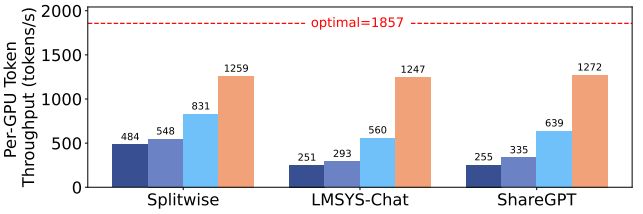
6 Evaluation

We now mainly answer the following questions:

- How much NanoFlow improves serving throughput compared to existing systems, and how does its throughput compare to the optimal serving throughput? (§6.2)
- What is the average serving latency under NanoFlow for different request rates, and how is it distributed? (§6.3)



(a) LLaMA-2-70B, 8 GPU, TP=8, Constant Input & Output Length



(b) LLaMA-2-70B, 8 GPU, TP=8, Input & Output Length from Dataset

Figure 7: Offline throughput comparison. NanoFlow outperforms all baselines for all the workload settings. TP stands for the number of GPUs used with tensor parallelism.

- How do the various techniques proposed in NanoFlow contribute to the end-to-end throughput? (§6.4)
- What is the compute, memory and network resource usage pattern of NanoFlow? (§6.5)
- How does NanoFlow improve performance when applied to different popular LLMs? (§6.6)

6.1 Experiment Setup

Hardware. We run our experiments on 8× A100 80GB SXM GPUs interconnected via NVLink.

Models. We provide a detailed evaluation of NanoFlow using the LLaMA-2-70B model, one of the most widely-used open-source LLMs. We also demonstrate the applicability of NanoFlow by evaluating it on 5 other representative LLMs. We use FP16 weights and activations for all models, as this is

Dataset	Avg. Input (Std)	Avg. Output (Std)
Splitwise [32]	1155 (1109)	211 (163)
LMSYS-Chat [56]	102 (169)	222 (210)
ShareGPT [1]	246 (547)	322 (244)

Table 4: The average and standard deviation of input and output lengths in the sampled datasets.

standard for data center-scale inference.

Baselines. We consider three widely-used serving frameworks as baselines.

vLLM[6] [17, 52] is a state-of-the-art serving system delivering high throughput. vLLM implements pagedAttention for increasing GPU memory utilization, as well as the chunked prefill for higher GPU utilization.

DeepSpeed-FastGen[7] [13, 23] is a serving framework developed by Microsoft. It dynamically composes prefill with decode requests to ensure that the engine is operating in a high throughput regime. We vary the `max-ragged-batch-size` to tune the batch size for highest throughput.

TensorRT-LLM[8] [26, 27] is a high-performance LLM inference engine built upon NVIDIA's TensorRT SDK. We set `max-num-tokens` by calculating the maximum capacity for the KV-cache in the GPU memory. We also enable paged KV-cache and dynamic batching optimizations when compiling.

Datasets. Splitwise [32] is a conversation trace collected from a real production environment at Microsoft, with a total of around 20000 requests. LMSYS-Chat-1M [56] is a large-scale dataset with 1 million real-world conversations from 25 different LLMs. ShareGPT [1] is a dataset with conversations collected from the ShareGPT API. We use the full trace from Splitwise and randomly sample 50,000 requests from ShareGPT and LMSYS-Chat-1M for our evaluation. Table 4 shows the average input length and output length in tokens for the sampled datasets we use.

6.2 Throughput

We first evaluate throughput to simulate use cases such as benchmarking, information extraction, data wrangling, and form processing [36]. We sample actual input (prompt tokens) and output (generated tokens) lengths from the datasets and add additional experiments with constant length. We then deploy a NanoFlow instance (with a dense batch size of 2048 in the case of Llama-2-70B, where NanoFlow delivers best performance) to measure the *total throughput* defined as the total number of input and output tokens divided by the execution time. We compare the performance of the NanoFlow instance with optimal throughput derived using Equation 5, where we measure the peak compute capacity using the state-of-the-art GEMM vendor library CUTLASS with a token batch size of 2048. Since the optimal throughput is independent of user query statistics, for all the experiments, the optimal throughput is 1857 tokens/s/GPU as computed in §3.5.

Figure 7 shows the throughput of the NanoFlow instance compared to the baselines. NanoFlow has the highest throughput in all settings and is able to achieve 68.5% of the theoretical optimal throughput in the best case. For constant lengths, NanoFlow has on average 2.62× higher offline throughput than vLLM, 2.78× higher than DeepSpeed-FastGen and 1.73× higher than TensorRT-LLM. When input and output lengths are drawn from the datasets, NanoFlow has on average 4.18× higher throughput than vLLM, 3.45× higher than DeepSpeed-FastGen and 1.91× higher than TensorRT-LLM.

6.3 Latency

We evaluate the latency of the NanoFlow instance, for which we model the request arrival interval via an exponential distribution, following prior work [17]. We generate 5 minutes of request traces from the datasets in Table 4 for various request rates. We then measure *normalized latency* by first calculating the end-to-end request latency divided by output length in tokens, then taking the average for all requests. For each framework, we gradually increase the request rate and measure the normalized latency. We select the SLO for average normalized latency as 200*ms* following prior works [59], which is also the typical human reading speed [8].

Figure 8 demonstrates the normalized latency of the NanoFlow instance compared to baselines at various request rates. At lower rates, the NanoFlow instance has comparable but slightly higher latency, because NanoFlow targets throughput-oriented scenarios and therefore employs a large dense batch size.

The NanoFlow instance is able to sustain a higher request rate within 200*ms* latency SLO compared to all other baselines across all the datasets. For example, for LMSys-Chat-1M dataset, the NanoFlow instance can achieve 1.64× higher request rate compared to TensorRT-LLM within 200ms normalized latency constraint.

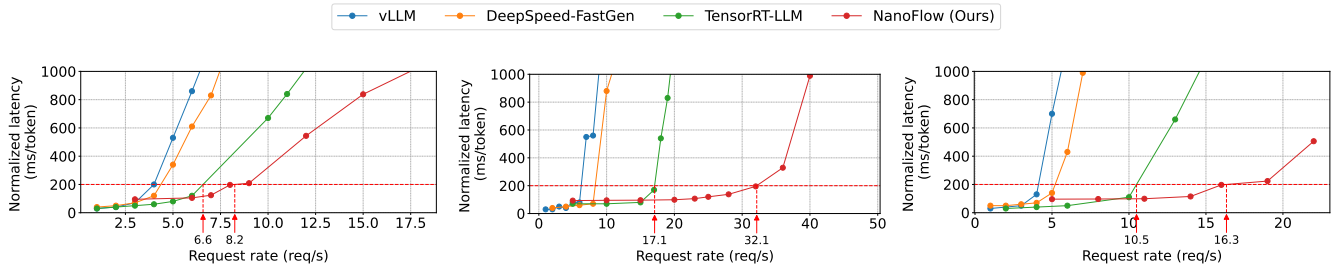
Moreover, the NanoFlow instance's 99th-percentile latency is only 1.07× of the average latency at near-maximum throughput, as NanoFlow uses a constant dense batch size, helping it perform well even under tail latency SLOs.

6.4 Ablation Study

To showcase the overhead and benefit of the techniques that NanoFlow uses, including nano-batching, nano-operation overlapping, and KV-cache offloading, we compare the NanoFlow instance with baselines that share the same asynchronous request scheduling and kernel libraries.

We created two baselines: a non-overlapping baseline processing inputs sequentially without nano-batches, and a nano-batch-only baseline that splits requests into nano-batches as in NanoFlow but executes them sequentially to assess nano-batching overhead. As shown in Figure 9, splitting into nano-batches alone reduces performance by 13.2%.

[6] v0.5.3.post1 (commit ID 38c4b7e)
[7] v0.2.3 (commit ID 429bc5c)
[8] v0.8.0 (commit ID 5955b8a)



(a) Splitwise (b) LMSYS-Chat-1M (c) ShareGPT

Figure 8: Latency comparison. The x-axis shows the number of incoming requests per second and the y-axis shows the normalized latency. NanoFlow handles higher request within 200ms SLO constraints.

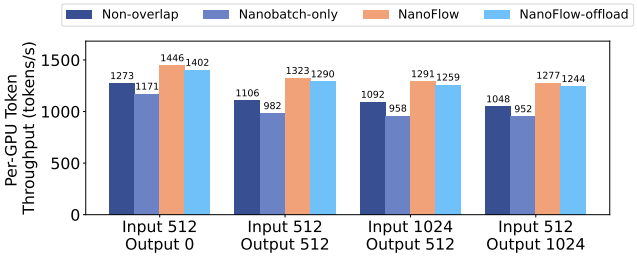


Figure 9: Ablation study results for NanoFlow. Nano-batching and overlapping improves NanoFlow's performance.

To estimate the benefit of overlapping network- and memory-bound kernels, we compare NanoFlow and baselines under various prefill-decode ratios, as shown in Figure 9. We use prefill-only workloads (Input 512, Output 0) to demonstrate the benefit of overlapping network-bound and compute-bound kernels. Additionally, we use decode heavy workloads (Input 512, Output 1024) to evaluate the benefit of overlapping both network- and memory-bound kernels. NanoFlow achieves 1.07× speedup by overlapping network-bound kernels and 1.17× speedup by overlapping both network- and memory- bound kernels compared with the non-overlapped baseline.

Moreover, we quantify the performance degradation due to offloading in Figure 9. Enabling offloading would slow down the pipeline by 3.0% due to kernel interference caused by KV-cache movement. However, offloading can reduce 3.02x compute for multi-round LMSYS-Chat workloads.

6.5 Resource Usage

We demonstrate NanoFlow's effectiveness in terms of resource utilization in Figure 10. While the non-overlapping baseline sequentially executes operations, which mostly uses only one resource at a given time, the NanoFlow instance can concurrently utilize multiple resources and achieves 68.5% average compute utilization. Due to kernel interference, NanoFlow provides lower than optimal compute usage.

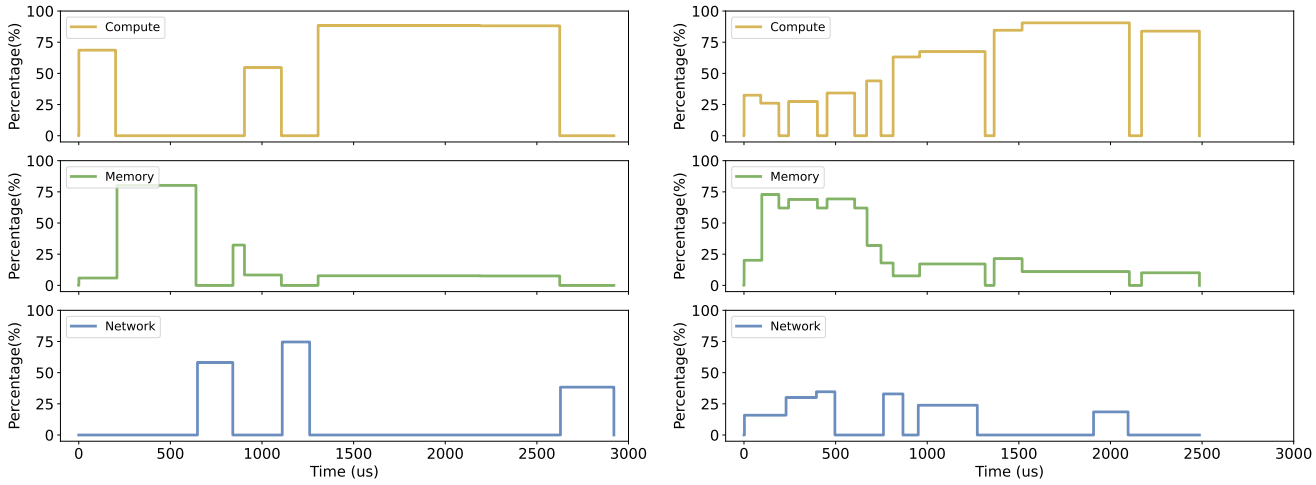
6.6 Performance on Other LLMs

We evaluate the throughput of NanoFlow instances for other LLMs with a constant length of input 1024 and output 512. All evaluations are done on 8×A100 80GB SXM GPUs, except for LLaMA-3-8B, which runs on a single A100 80GB SXM. The results are summarized in Figure 11. We find that NanoFlow improves throughput to between 50% and 72% of optimal throughput for these models, surpassing vLLM.

7 Related Work

LLM serving optimizations. Prior works have investigated optimizations for improving the serving throughput at different granularities. Orca [53] proposes request-level continuous batching, which refills the on-the-fly batch to maximize batch size at the granularity of an iteration. DistServe [59] and Splitwise [32] explore phase-level scheduling, which disaggregates the prefill and decode phases into different clusters. DeepSpeed-FastGen [13] and Sarathi-Serve [2] propose a chunked prefill policy, which splits a prefill request into multiple smaller chunks and batches decode and prefill requests together to improve overall utilization. However, these works do not perform scheduling at the granularity of operations. NanoFlow exploits intra-device parallelism, which features fine-grained resource management. Other works tackle the inference inefficiency by optimizing specific operations, including PageAttention [17], quantization approaches [19, 44, 55], etc. NanoFlow can easily adopt new operation implementations by profiling their performances and interference charateristics and integrating them to auto-search.

Operation-level parallelism. Some works focus on improving the efficiency of DNN workloads with operation-level optimizations. For example, Rammer [20] explores intra-operation parallelism by remapping the operations into different functional units. Unity [50] investigates the combination of parallelism and equivalent algebraic transformation of operations. However, these works do not consider the different resource demands of operations. ASPEN [30] and Welder [37] break the boundary between operations by constructing and compiling a tile-level data graph. Since tiles follow sequential dependencies, parallelism is still limited. Moreover, prior



(a) Non-overlap pipeline resource usage

(b) NanoFlow resource usage

Figure 10: Non-overlapped pipeline and NanoFlow's resource usage during inference of a single layer. NanoFlow achieves high compute usage across the whole pipeline by simultaneously also utilizing memory and network bandwidth.

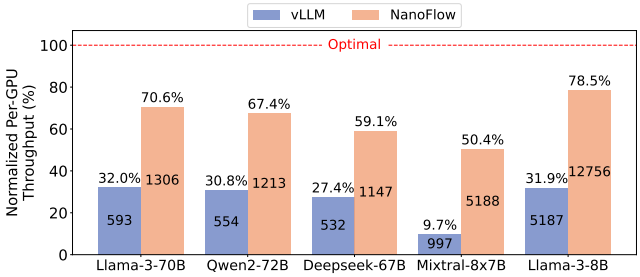


Figure 11: Performance of NanoFlow instances in terms of tokens per second per GPU for other models. Compared with vLLM, NanoFlow significantly increases throughput. NanoFlow achieves up to 78.5% of optimal throughput.

works require recompilation from scratch, which requires significant manual work. Compared with them, NanoFlow creates more overlapping opportunities through nano-batching.

8 Conclusion

In this work, we analyzed the characteristics of different operations in LLM serving and revealed the compute-bound nature of modern LLM serving workloads. We identified that compute is underutilized inside a single GPU due to the sequential execution of compute-bound, memory-bound, and network-bound operations, leading to sub-optimal throughput. To address this, we proposed NanoFlow, a novel end-to-end LLM serving system that overlaps operations with heterogeneous resource usages through intra-device parallelism. NanoFlow automatically constructs a pipeline that overlaps nano-operations, with an auto-search adaptive to various LLM models. Our experiments show that NanoFlow achieves 1.91× throughput improvements over the state-of-the-art systems.

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